

TrES-3b

R-1411

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160327 · created 2026-08-01T05:15:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457474.8902 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.023
Transit depth (Rp/Rs)^2	3.8 +/- 0.88 [%]
Orbital Inclination (inc)	81.76 +/- 1.08
Airmass coefficient 1 (a1)	1.032 +/- 0.016
Airmass coefficient 2 (a2)	-0.0199 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#3 - [114, 217]
Optimal Aperture	3.34
Optimal Annulus	11.41
Transit Duration (day)	0.056 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.023 vs 0.16309 (deviation 1.39σ)
Duration fit vs published	0.056 d vs 0.0567 d (deviation 0.06σ)
Mid-time O–C	2.1 min
Depth SNR	8.5
Residual scatter	1.55 %

Light curve

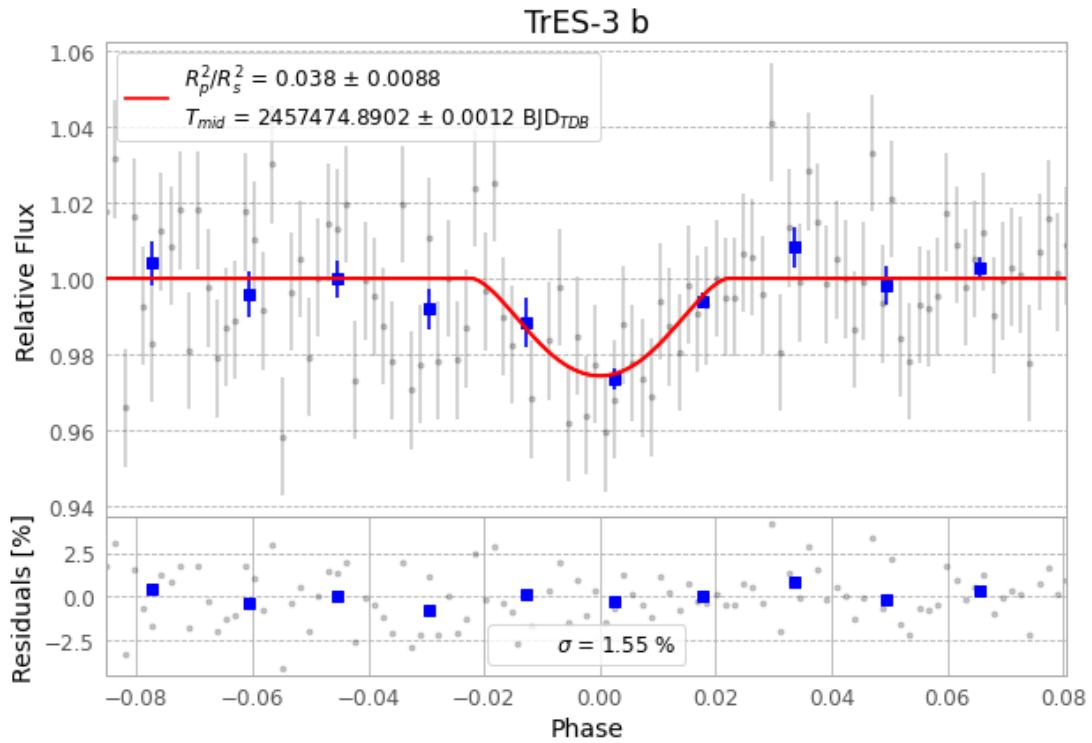


Figure 1 — FinalLightCurve_TrES-3 b_2016-03-26.png (SHA-256 83afd471ecf54f1e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 124], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50c6c9654c80807e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

105 FITS frames (69 MB), TRES-3160327063912.FITS → TRES-3160327115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160327.log (526b6113e53c...), FinalLightCurve_TrES-3 b_2016-03-26.png (83afd471ecf5...), FinalLightCurve_TrES-3 b_2016-03-26.csv (d2f8369ec63b...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:15Z.